

Overworked Americans

Problem 1: Q1, Q4, Q5, Q7

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Q1 — Documenting the EU–US hours gap

Q1 — Question and roadmap

“Document the EU–US (and intra-EU) hours-worked gap in recent years.”

What we deliver.

- ❖ One stylised fact (three panels): hours fell, the tax wedge does not cleanly explain the cross-section.
- ❖ Two questions panel (c) raises:
 - ❖ (I) the true slope of the tax–hours relation;
 - ❖ (II) the wide *residual scatter* around it ($\sigma_{\text{resid}} \approx 158$ hours).
- ❖ Q4 (cap) and Q5 (union bargaining) develop two institutional channels that sit in those residuals as endogenous confounders.

Decomposition. For the working-age (15–64) population, $H/N = HE \cdot e$. Bick–Brüggemann–Fuchs–Schündeln (2019) attribute roughly half of the EU–US gap to each margin. We focus on HE (intensive margin) — the prompt asks about hours.

Q1 — The fact

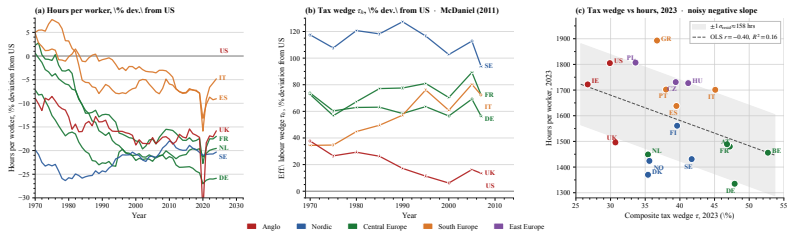


Figure 1: (a) Hours per worker, % deviation from US, 1970–2024. (b) Effective labour wedge τ_h , % deviation, 1970–2007. (c) OECD labour-tax wedge vs hours per worker, 2023 cross-section ($n=19$); OLS line and $\pm 1\sigma_{\text{resid}}$ band.

Q1 — Three readings of the panel

- ❖ (i) **Hours fell relative to US since 1970:** DE/NL ≈ -25 to -30% ; IT/ES ≈ -10 to -15% .
- ❖ (ii) **Wedge cross-country pattern is unstructured:** Italy's relative wedge drifts up, the UK's narrows, France stays flat. No clean mirroring of panel-(a) hours.
- ❖ (iii) **The 2023 cross-section is signed correctly but noisy:** OLS slope -9.9 hrs/pp, $R^2=0.16$, residual SD ≈ 158 hours $\approx$ a third of the entire DE-US gap. Italy (45.1%) and Greece (36.7%) work *more* than France (46.8%); the four highest- τ countries (BE, DE, AT, FR) span ≈ 200 hours within a 6 pp wedge band.

Bottom line. Tax wedges sign the relation correctly but cannot pin its slope (endogenous τ) and cannot account for the scatter. Q4 and Q5 are not orthogonal additions to the tax effect: they live in the residuals as endogenous confounders.

Q4 — Hours regulation: optimality of a cap

Q4 — Setup: three bliss points and the firm-power ordering

Primitives (quadratic-disutility worker, linear-quadratic firm):

$$V_w(h) = (1-\tau)wh - \frac{\alpha}{2}h^2, \quad V_f(h) = Ah - \frac{\beta}{2}h^2 - wh.$$

Wage transfer cancels in $W = V_w + V_f$. Three FOCs give three bliss points:

$$h^* = \frac{(1-\tau)w}{\alpha}, \quad h^{\text{soc}} = \frac{A - \tau w}{\alpha + \beta}, \quad h^{\text{firm}} = \frac{A - w}{\beta}.$$

Cap as quantity instrument. The firm picks h to maximise V_f s.t. $h \leq \bar{h}$, with w pinned by IR $V_w(h) = V_w^0$. KKT yields $h^{\text{eq}}(\bar{h}) = \min\{\bar{h}, h^{\text{firm}}\}$.

Firm-power ordering. At baseline $(A, w, \tau, \alpha, \beta) = (2, 1, 0.4, 2, 2)$:

$$h^* = 0.30 < h^{\text{soc}} = 0.40 < h^{\text{firm}} = 0.50.$$

The firm overshoots the social optimum: a binding cap has something to correct.

Q4 — Optimality condition for the cap

Three surpluses (relative to the no-cap baseline h^{firm}):

$$\begin{aligned}\Delta V_f(\bar{h}) &= -\frac{\beta}{2} (h^{\text{firm}} - \bar{h})^2 \leq 0, \\ \Delta W(\bar{h}) &= -\frac{\alpha+\beta}{2} \left[(h^{\text{soc}} - \bar{h})^2 - (h^{\text{soc}} - h^{\text{firm}})^2 \right].\end{aligned}$$

Welfare condition (W is symmetric in $\bar{h}$ around h^{soc}):

$$\Delta W(\bar{h}) > 0 \iff \bar{h} \in (2h^{\text{soc}} - h^{\text{firm}}, h^{\text{firm}}).$$

Peak at $\bar{h} = h^{\text{soc}}$ with gain $\frac{\alpha+\beta}{2} (h^{\text{firm}} - h^{\text{soc}})^2 = 0.020$ at calibration.

Optimality condition. The cap is welfare-improving *iff* it pulls hours toward h^{soc} without overshooting; it destroys joint surplus when $\bar{h} < 2h^{\text{soc}} - h^{\text{firm}}$ (firm losses dominate). At $\alpha = \beta$ the lower band edge collapses to $h^* = 0.30$.

Q4 — Simulation: welfare gain over the cap

Hours cap: optimal at $\bar{h} = h^{\text{soc}}$; welfare-improving on $\bar{h} \in (2h^{\text{soc}} - h^{\text{firm}}, h^{\text{firm}})$, overshoot below

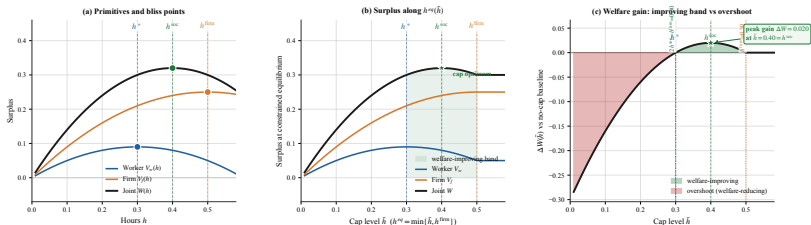


Figure 2: (a) Primitives V_w , V_f , W over h with bliss points marked. (b) Surpluses along $h^{\text{eq}}(\bar{h}) = \min\{\bar{h}, h^{\text{firm}}\}$. (c) $\Delta W(\bar{h})$ vs no-cap baseline; **green** = welfare-improving band, **red** = overshoot. Peak gain $\Delta W = 0.020$ at $\bar{h} = h^{\text{soc}} = 0.40$.

Q4 — Two thresholds and the Aubry anchor

Two distinct thresholds as the cap tightens past h^{soc} :

- ❖ Joint surplus turns negative once $\bar{h} < 2h^{\text{soc}} - h^{\text{firm}} = 0.30$ (firm losses dominate worker gains).
- ❖ The individual worker still benefits until $\bar{h} = h^* = 0.30$, and only becomes worse off than no-cap at $\bar{h} < 2h^* - h^{\text{firm}} = 0.10$.
- ❖ For $\bar{h} \in (0.10, 0.30)$: *cap reduces aggregate welfare but the worker still gains.*

Empirical anchor (Aubry, France 1998–2000). Estevão-Sá (2008): $\approx 10\text{--}13\%$ hours cut, near-zero employment effect. Consistent with a binding cap at $\bar{h}/h^{\text{firm}} \approx 0.875$, well *inside* the welfare-improving band.

Q5 — A single union as welfare-restorer

Q5 — Right-to-manage Nash bargain

Setup. Single encompassing union (Calmfors–Driffill 1988); on the firm's labour demand $h_j^d(w) = (A_j - w)/\beta$, with $x_j \equiv A_j - w$:

$$V_w(x_j) = A_v x_j - B_v x_j^2, \quad V_f(x_j) = \frac{x_j^2}{2\beta},$$

$$A_v \equiv (1-\tau)A_j/\beta, \quad B_v \equiv (1-\tau)/\beta + \alpha/(2\beta^2).$$

Bargain. Union and firm pick w via asymmetric Nash with worker weight $\eta \in [0, 1]$:

$$\max_w [V_w(w; A_j) - V_w^0]^\eta V_f(w; A_j)^{1-\eta}.$$

Interior FOC reduces to a quadratic in x_j :

$$2B_v x_j^2 - A_v(2 - \eta) x_j + 2(1 - \eta) V_w^0 = 0.$$

Larger root $x_j^*(\eta)$ delivers $w_j(\eta) = A_j - x_j^*$, $h_j(\eta) = x_j^*/\beta$.

Boundaries. $\eta=0$: IR binds ($V_w = V_w^0$), $h_j = h_j^{\text{firm}}$. $\eta \rightarrow 1$: $x_j^*(1) = A_v/(2B_v)$, $h_j(1) < h^*(w_j(1))$.

Q5 — Comparative statics and the welfare-restorer

Implicit differentiation of the bargain quadratic in η :

$$\left[4B_v x_j - A_v(2 - \eta) \right] \frac{dx_j}{d\eta} = -A_v x_j + 2V_w^0.$$

Bracket on LHS = $\sqrt{D(\eta)} > 0$ on $[0, 1]$ under (R1) $A_v^2 > 4B_v V_w^0$. RHS strictly negative since $A_v x_j^* \geq A_v^2 / (2B_v) > 2V_w^0$ (also (R1)). So:

$$\frac{dx_j}{d\eta} < 0, \quad \frac{dw_j}{d\eta} > 0, \quad \frac{dh_j}{d\eta} < 0.$$

Welfare-restorer. W concave with peak at $h_j^{\text{soc}} \in (h_j^*, h_j^{\text{firm}})$. Since $h_j(\eta)$ is continuous and strictly decreasing with $h_j(0) = h_j^{\text{firm}}$, IVT gives a unique $\hat{\eta}_j \in (0, 1)$ with $h_j(\hat{\eta}_j) = h_j^{\text{soc}}$:

$$W(h_j(\eta)) - W(h_j^{\text{firm}}) = -\frac{\alpha + \beta}{2} \left[(h_j^{\text{soc}} - h_j(\eta))^2 - (h_j^{\text{soc}} - h_j^{\text{firm}})^2 \right]$$

increasing on $[0, \hat{\eta}_j]$, decreasing thereafter (overshoot logic of Q4).

Calibration. OECD ICTWSS coverage maps US ($\sim 12\%$) to $\eta^{US} \approx 0.15$; EU ($\sim 60\text{--}70\%$) to $\eta^{EU} \approx 0.5$. EU–US hours gap ≈ 0.08 ($\approx 19\%$ logs).

Q5 — Sectoral shocks: pass-through and asymmetry

Asymmetric pass-through: good shocks $\rightarrow$ wage rents (hours fall, profit roughly preserved); bad shocks $\rightarrow$ amplified hours and profit collapse under wage rigidity

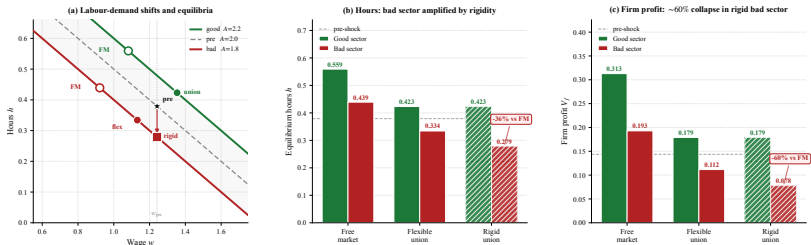


Figure 3: Mean-preserving sectoral shock $A_g = 2.2$, $A_b = 1.8$ at $\eta = 0.5$. (a) Demand shifts and equilibria; rigid bad-sector at stuck w_{pre} . (b) Hours: bad-sector rigid is 36% below FM. (c) Firm profit: bad-sector rigid is 60% below FM.

Q5 — Asymmetric pass-through under wage rigidity

Flexible union ($A_g=2.2$, $\eta=0.5$). $w_g^U=1.354$, $h_g^U=0.423$. Productivity gain splits into wages ($\Delta w/w=+25\%$) and reduced hours ($\Delta h/h=-24\%$); joint surplus falls only $\approx 5\%$ ($W_g^{FM}=0.363 \rightarrow W_g^U=0.344$). **Bulk of redistribution = wage-rent transfer, not deadweight loss.**

Rigid bad sector ($A_b=1.8$, w stuck at $w_{pre}=1.242$). Hours read off labour demand at the stuck wage:

$$h_b^{rig} = (A_b - w_{pre})/\beta = 0.279.$$

Hours -36% vs FM, profit -60% vs FM.

Headline.

- ❖ Good shocks pass through *diluted*: absorbed in wage rents.
- ❖ Bad shocks pass through *amplified*: rigidity forces hours to absorb the entire shock.

The encompassing-union setup + downward wage rigidity makes the channel asymmetric across positive and negative shocks — a feature absent from both the Walrasian benchmark and the Q4 cap.

Q7 — Welfare comparison: a revealed-preference reading

Q7 — λ -weighted welfare and the optimum locus

Whose welfare? Saez–Stantcheva (2016) generalised welfare weight: planner attaches premium $\lambda \in [0, 1]$ to a worker dollar over a firm-owner dollar:

$$W^\lambda(\eta) = (1 + \lambda) V_w(\eta) + (1 - \lambda) V_f(\eta).$$

$\lambda = 0$: equal-weight utilitarian. $\lambda = 1$: Rawlsian. Following Alesina–Angeletos (2005), λ is a country-specific *fairness primitive*.

Closed form (at baseline). Maximising W^λ along the bargained path of Q5:

$$x^*(\lambda) = \frac{0.3 A (1 + \lambda)}{0.6 + 1.6 \lambda}, \quad \eta^*(\lambda) = \frac{1.2 x^* - 1.1 (x^*)^2 - 0.1}{0.6 x^* - 0.1}.$$

Boundaries: $\eta^*(0) = 0$ (firm-power optimal under equal weights), $\eta^*(1) = 1$ (full union under Rawlsian); η^* monotonically increasing in λ .

Two readings. Map observed coverage $\hat{\eta}^{US} \approx 0.15$, $\hat{\eta}^{EU} \approx 0.50$ to either: (A) both countries share a common λ , or (B) each country at-optimum given its own country-specific λ^{rev} .

Q7 — Reading A: common λ

Anchor. HSV (2017) progressivity index $\tau_p \approx 0.18$ maps under log preferences to $\lambda \approx 0.30$ on the Saez–Stantcheva scale.

Implication. Welfare-optimum $\eta^*(0.30) = 0.58$. Both observed $\hat{\eta}$ are *below* the optimum:

US ($\hat{\eta} = 0.15$): under-unionised by 0.43; EU ($\hat{\eta} = 0.50$): mildly under by 0.08.

Welfare losses $W^{0.30}(\eta^*) - W^{0.30}(\hat{\eta})$:

- ❖ US: $0.282 - 0.259 = \mathbf{0.022}$ ($\sim 8\%$ of joint surplus on the table).
- ❖ EU: $0.282 - 0.281 = \mathbf{0.001}$ (essentially at-optimum).

Verdict under Reading A. EU institutions are essentially welfare-optimal under any reasonable λ ; US institutions leave $\sim 8\%$ on the table because the firm extracts too much. The mechanism is institutional implementation, not preferences.

We flag this reading but do not endorse it (see Reading B).

Q7 — Reading B: country-specific λ (preferred)

Inverting $\eta^*(\lambda)$ on observed coverage:

$$\hat{\eta}^{US} = 0.15 \implies \lambda^{US, \text{rev}} \approx 0.05, \quad \hat{\eta}^{EU} = 0.50 \implies \lambda^{EU, \text{rev}} \approx 0.24.$$

Each country at its own optimum ($\Delta\lambda \approx 0.19$):

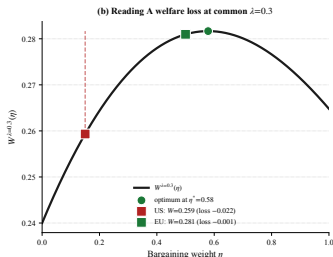
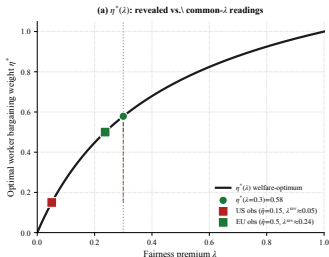
- ❖ US: $h^{\text{firm}} \rightarrow 0.46$ (−7%). Small union-channel hours drag.
- ❖ EU: $h^{\text{firm}} \rightarrow 0.38$ (−24%). Large union-channel hours drag.

Verdict under Reading B. Neither country is welfare-better in absolute terms — each implements its own optimum given its own fairness primitive. The EU–US hours gap reflects a *fairness-belief gap*, not institutional misalignment.

Why Reading B is preferred. Reading A would put *all* countries on the horizontal line $\eta = \eta^*(0.30) = 0.58$. The observed range (US 0.15, FR 0.98, SE 0.88, DE 0.52) clearly violates that prediction.

The missing variable in Q1 panel (c) is reconstructed by the λ -driven union channel: the $\Delta\lambda \approx 0.19$ gap is precisely the kind of unobserved heterogeneity the noisy tax–hours OLS cannot pin down.

Q7 — Diagnostic and final verdict



Verdict. Reading B accommodates observed cross-country bargaining heterogeneity; Reading A cannot. **The cross-country hours gap is a fairness-belief gap, recovered from observed bargaining institutions + Q4/Q5 model structure.**

Synthesis

Synthesis: what the four channels deliver together

The chain.

- ❖ **Q1.** Tax wedges sign the EU–US hours gap correctly but leave a residual SD ≈ 158 hours — a third of the entire DE–US gap.
- ❖ **Q4.** A binding hours cap pulls h from h^{firm} toward h^{soc} . Optimal at $\bar{h} = h^{\text{soc}}$; welfare-improving on $(2h^{\text{soc}} - h^{\text{firm}}, h^{\text{firm}})$. *Quantity instrument.*
- ❖ **Q5.** An encompassing union with bargaining weight η delivers the same overshoot logic via wages: $dh/d\eta < 0$, hump-shaped joint surplus, asymmetric pass-through under wage rigidity. *Price instrument.*
- ❖ **Q7.** Confronting Q5 with US/EU coverage data + Saez–Stantcheva λ recovers a fairness-belief gap $\Delta\lambda \approx 0.19$ — the missing variable in Q1's noisy cross-section.

What the comparative structure surfaces. A result no single-channel treatment could produce: each country's institutional configuration is rationalised as the at-optimum response to its own fairness primitive. The EU–US hours gap is not an institutional misalignment to be corrected, but the equilibrium expression of cross-country preference heterogeneity.

Thank you

Questions?

Full paper, companion appendix, replication scripts:

github.com/AlessandroCaggia30/overworked-americans-takehome